COMMONWEALTH OF AUSTRALIA

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Southern Cross University

Pharmacology of the Respiratory System

Introductory Pharmacology
PHAR2001

Dr Ken Wojcikowski

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Objectives for the whole system

Southern Cross University

- **Briefly describe the structure and function of the lungs including the immune system response in the lungs**
- **Briefly describe the pathophysiology of important lung disorders, with an emphasis on:**
  - **allergic rhinitis and**
  - **asthma**
- **...and for those two disorders, briefly identify and describe the actions of common pharmacological agents used in their treatment**
- **Compare the difference between mucolytic agents and expectorants, citing examples of each.**

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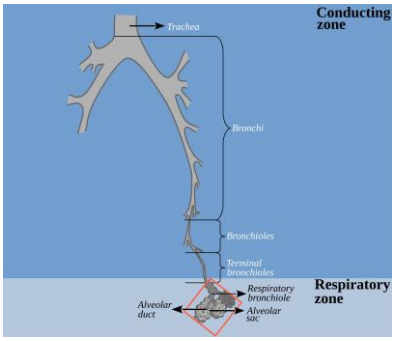
Medications of the Respiratory System:

Lesson 1

Review of the Respiratory System

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Structure of the pulmonary system



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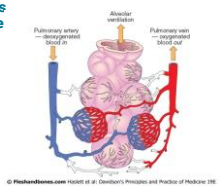
Actual size of lungs



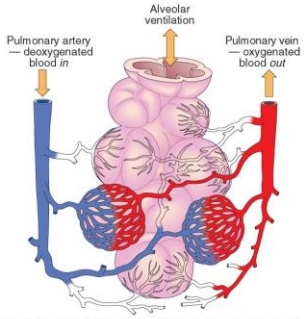
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Structure of the Respiratory System

- Consists of the air passages and the lungs
- Divided into two parts by function:
  - **Conducting airways,** through which air moves as it passes between the atmosphere and the lungs
  - **Respiratory tissues of the lungs,** where gas exchange takes place



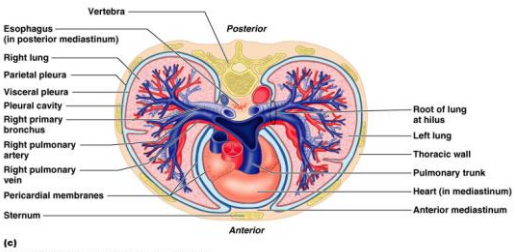
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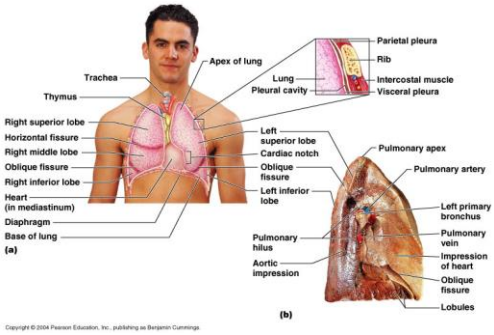
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Lung Circulation Involves Two Separate Circulations



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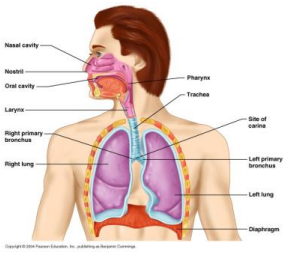
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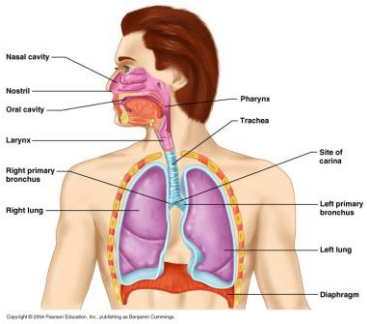
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Ventilation

- **Bulk flow of air**
- **Disorders that obstruct ventilation are called obstructive disorders**
  - **Asthma, COPD, others**
- **Function**
  - **Moves air out of the lungs but does not participate in gas exchange**



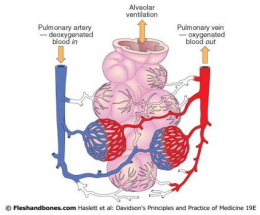
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External Respiration (Also referred to as gas exchange)

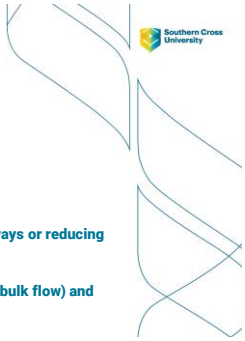
- Exchange of gases from an area of high concentration to an area of lower concentration
- Takes place in the terminal respiratory bronchioles and the alveolar ducts and sacs
- Can be reduced with inflammation, fibrosis, or decreased area



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How can pharmaceutical agents help?

- Fighting infection
  - Antibiotics, antifungal agents
- Preventing excess mucus production
- Decreasing cough
- Expelling mucus
- Promoting ventilation or 'bulk flow of air'
  - Promote airway adequacy by dilating airways or reducing excess mucus
- Reducing inflammation
  - Inflammation can affect both ventilation (bulk flow) and respiration (exchange of gasses).



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Medications of the Respiratory System:

Lesson 2

Antitussives, expectorants and mucolytic agents

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Antitussives (suppress cough reflex)

Cough suppressants

- For Non-productive cough only
  - Depress the cough reflex located in the medullary cough centre (centrally acting)
  - Codeine (this is an opioid narcotic),
  - Dextromethorphan (this is a morphine derivative but not considered an opioid)

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Antitussives (suppress cough reflex)

Cough suppressants

- Codeine: Adverse effects:
  - Drowsiness, sedation (CNS depression), respiratory depression
- Dextromethorphan: Adverse effects:
  - Allergic reactions, GIT complaints
- Contraindications for codeine:
  - Pregnancy, young children, especially under 2
- Dextromethorphan is a pregnancy category A drug

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Expectorants

- Stimulate mucus secretion
- Reduce the viscosity of thick tenacious
- Guaifenesin (in Benadryl) PO
- May result in some GI symptoms

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Mucolytic agents

- For productive coughs
  - Chronic bronchitis, pneumonia, cystic fibrosis
- Decreases mucus viscosity/ liquefies secretions
  - Facilitates airway and secretion clearance
- Acetylcysteine (know this drug)
  - Breaks down mucoproteins
  - Acetylcysteine is also the antidote for paracetamol toxicity
- Dornase alfa
  - Breaks down DNA from proteins (specifically for cystic fibrosis)
- Adverse effects
  - Pharyngitis, GI disturbances

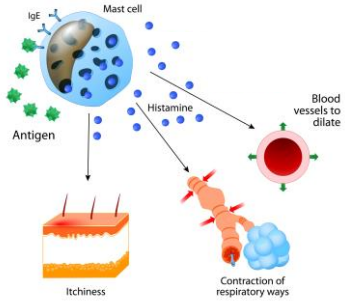
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Medications of the Respiratory System:
Lesson 3
Decongestants and Antihistamines

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In allergic rhinitis, histamine dilates the blood vessels of the sinuses.

Decongestants constrict those blood vessels by acting on the alpha 1 adrenergic receptors.



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The sinuses

Normally, the sinuses are empty except for a thin layer of mucus.

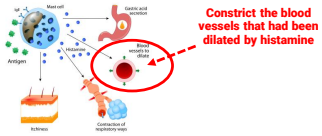
We probably have sinuses to humidify the air



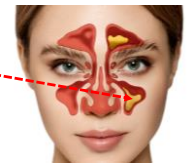
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Decongestants

- **Decongestants**
  - **Help stop 'runny nose'**
  - **Alpha agonists (act like adrenaline on the alpha 1 receptors)**
    - **Vasoconstriction = decrease in secretions, reduction in swelling of mucus membranes and oedema , opening of nasal passage**



Constrict the blood vessels that had been dilated by histamine



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Decongestants

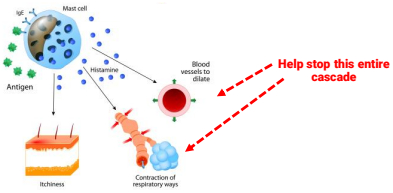
- **Nasal sprays**
  - **Oxymetazoline (Drixine)**
- **Oral medications**
  - **Pseudoephedrine (sale is restricted)**
  - **Phenylephrine (no better than placebo in most trials)**
- **Adverse effects**
  - **Rebound congestion (!: Don't use long-term)**
  - **Hypertension (other blood vessels are constricted)**
  - **Insomnia**



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Antihistamines (H1 antagonists)
Discussed in the tutorial

- H1 antagonists (NOT the H2 antagonists in gut)



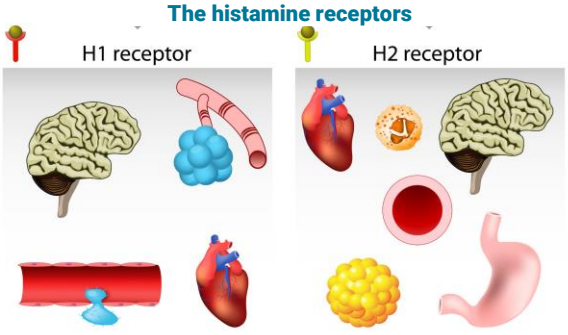
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Antihistamines (H1 antagonists)
Discussed in the tutorial

- H1 are first generation (cross BBB; Dimenhydrinate)
  - Side effects include drowsiness and decreased motion sickness
    - These are often used for their 'side effects'
  - These are associated with higher risk of Alzheimer's disease
- H1 can be second generation
  - do not cross BBB to the same degree, so less drowsiness
  - Loratadine and fexofenadine
- Decrease the vasodilation effects of histamine
- Antihistamines have varying amounts of secondary antimuscarinic actions
  - This is usually more profound with the first generation medications: dry mouth, difficulty urinating, eye pain

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The histamine receptors



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Medications of the Respiratory System:

Lesson 4

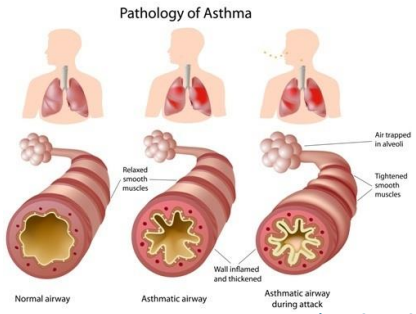
Overview of Lower Respiratory System Disorders

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Ventilation (obstructive) disorders

- COPD (emphysema **and** chronic bronchitis)
  - Other disorders chronically obstruct but are not COPD
- Chronic bronchitis (only)
- Emphysema (only)
- Asthma (10% of Australians have asthma)
- Cystic fibrosis (a fairly common genetic disorder)

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Decreasing the lumen's diameter by ½ = 1/16th the flow of air

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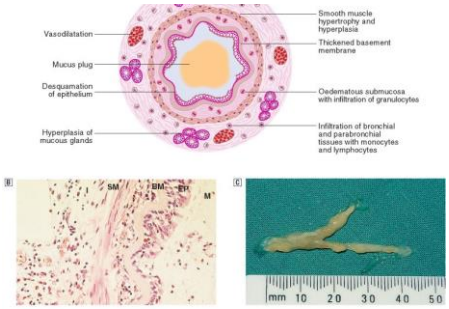
Asthma

- A condition characterised by:
  - Bronchospasm
  - Inflammation
  - Hyperactive airways
    - Mucus hyper secretion
    - Oedema
- Extrinsic
  - Allergens
- Intrinsic
  - Non-allergic: anxiety, exercise



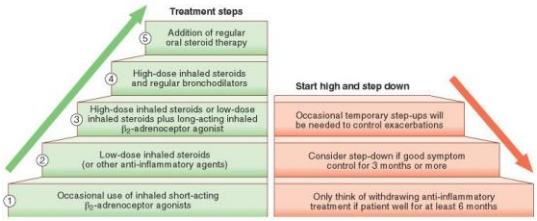
It may be best to consider an asthmatic to have a cup. Everything that irritates THEIR asthma fills up the cup. When the cup overflows, they have an asthmatic attack.
Try to get them to eliminate anything that contributes to their asthma.

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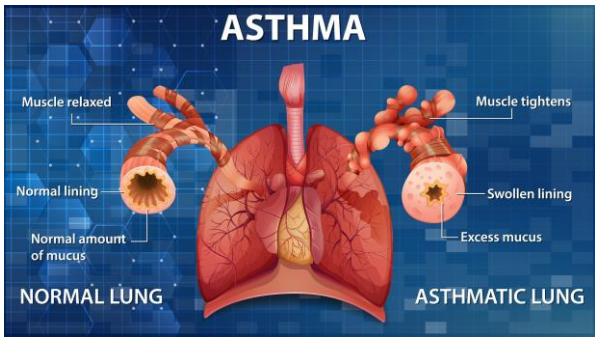


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Medications of the Respiratory System:

Lesson 5

Treatment of Asthma (and other ventilation disorders)

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Treatment of Asthma

- Antihistamines (PO; oral; efficacy is questioned)
- Bronchodilators (MDI; metred-dose inhaler)
  - Beta 2 agonists
  - Anticholinergic drugs
  - Others, for instance xanthines (oral)
- Corticosteroids (MDI)
- Leukotriene inhibitors (PO)
- Corticosteroids (PO)

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Antihistamines in asthma

- Antihistamines
  - Block the release of histamine
    - Reduces one mediator in the inflammatory and allergic response
    - Limited use in asthma because the mast cell that releases histamine also releases other stronger mediators of inflammation and the asthmatic is normally more sensitive to those. But some asthmatics respond.
  - Adverse effects mainly seen with first generation antihistimines (the ones that cause drowsiness)
    - Anticholinergic effects if using first generation: don't!
    - Dry mouth and drowsiness if using first generation: don't!
    - CNS depressants if using first generation: don't!
  - Use second generation antihistamines: Loratadine and fexofenadine and discontinue if no effect.

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Bronchodilators: inhaled

Dilate the airways, either mimicking the SNS (fight or flight) or blocking the PNS (rest and digest) in lungs



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Bronchodilators: inhaled

- Beta $_2$ agonist (mimics the adrenaline on the Beta $_2$ receptor on lungs)
  - Beta $_2$ receptors are located on the smooth muscle of blood vessels, bronchi and MAST cells
  - Adverse effects
    - Headache, Tachycardia after large systemic doses
  - Beta $_2$ selective agonists are long acting or short:
    - Salbutamol (Ventolin) (short acting rescue remedy)
    - Salmeterol (long acting, take with inhaled corticosteroid)



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Bronchodilators

- Anticholinergics/ antimuscarinic (against the parasympathetic nervous system's main neurotransmitter, acetylcholine)
  - Block acetylcholine (muscarinic) receptors responsible for causing bronchoconstriction: allows the SNS to dominate
    - Bronchodilation
  - Adverse effects
    - Anticholinergic effects
      - Dry mouth, dizziness
  - Ipratropium bromide (Atrovent) (inhaler, nebuliser)
  - Interactions
    - Other anticholinergic drugs

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Leukotriene receptor antagonists

- Leukotrienes are inflammatory mediators
- Oral dose (PO; (means PER OS; Oral)) normally in addition to inhaled corticosteroids
- Antagonists
  - Block leukotriene action
    - Reduced inflammatory response
    - Reduction in oedema, mucus secretion, smooth muscle contraction
  - Contraindication
    - Liver impairment (metabolised by the liver)
    - Pregnancy
  - Adverse effects:
    - Diarrhoea, nausea (Absorbed rapidly in the GI tract)
  - Zafirlukast (Accolate) and Montelukast

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Corticosteroids

Inhaled corticosteroids

- Should be taken when the asthmatic starts reaching for their rescue inhaler more than 3 times per week
  - Decrease the inflammatory response
    - Prevent the rupture of mast cells > decreased chemical mediator response > reduced inflammatory and immune response = reduction in oedema, bronchoconstriction, mucus production = increased airflow
  - Adverse effects
    - Pharyngeal, laryngeal fungal infections (thrush)
  - Budesonide (Pulmicort)
  - Fluticasone (Seretide)

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Corticosteroids

- Oral corticosteroids
  - Suppress immune and inflammatory response
    - Reducing the formation of prostaglandins and leukotrienes (required for inflammatory response)
    - Inhibit and impair the function of WBCs
  - Adverse effects
    - Infection, hyperglycaemia
  - Prednisone (PO)
  - *Discussed in the endocrine system*



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References

- Haslett C, et al. Davidson's Principles and Practice of Medicine, 19th ed. Churchill Livingstone, 2003
- Marieb E. Human Anatomy and Physiology, 5th ed. Benjamin Cummings, San Francisco, 2001
- Martini FH, et al. Visual Anatomy and Physiology 1st ed. Benjamin Cummings, 2011
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